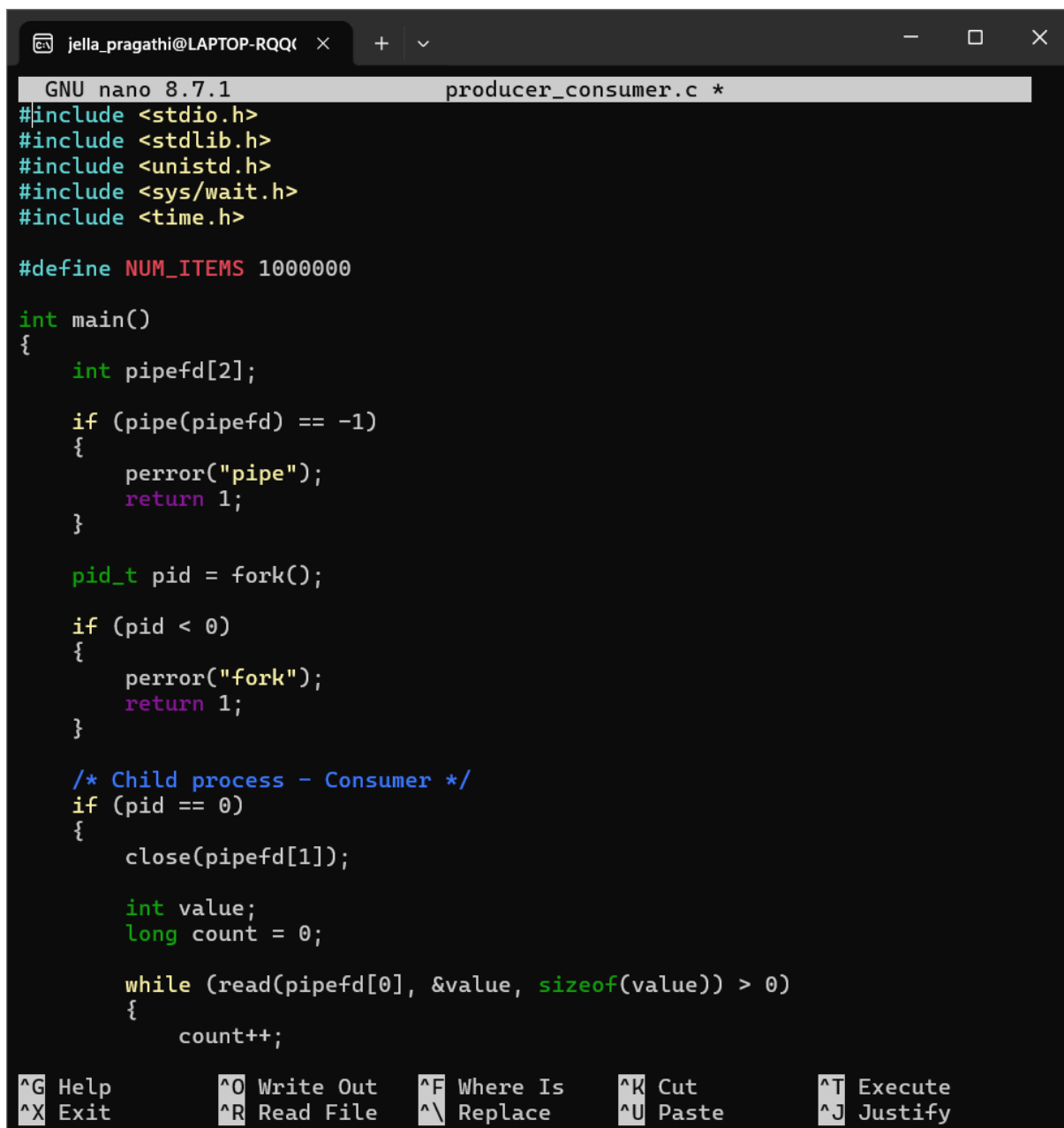


Practical week 5

Session 5

Task 1:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.



```
GNU nano 8.7.1 producer_consumer.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

#define NUM_ITEMS 1000000

int main()
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    /* Child process - Consumer */
    if (pid == 0)
    {
        close(pipefd[1]);

        int value;
        long count = 0;

        while (read(pipefd[0], &value, sizeof(value)) > 0)
        {
            count++;
        }
    }
}
```

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GNU nano 8.7.1 producer_consumer.c *

```
{
    count++;
}

close(pipefd[0]);

printf("\nConsumer Process\n");
printf("Child PID: %d\n", getpid());
printf("Items consumed: %ld\n", count);

exit(0);
}

/* Parent process - Producer */
else
{
    close(pipefd[0]);

    struct timespec start, end;

    clock_gettime(CLOCK_MONOTONIC, &start);

    for (int i = 0; i < NUM_ITEMS; i++)
    {
        int value = i;

        if (write(pipefd[1], &value, sizeof(value)) != sizeof(value))
        {
            perror("write");
            break;
        }
    }

    close(pipefd[1]);

    wait(NULL);
}
```

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```

        clock_gettime(CLOCK_MONOTONIC, &end);

        double elapsed =
            (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        printf("\nProducer Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Items produced: %d\n", NUM_ITEMS);
        printf("Communication time: %.6f seconds\n", elapsed);

        double data_mb =
            (NUM_ITEMS * sizeof(int)) / (1024.0 * 1024.0);

        double efficiency = data_mb / elapsed;

        printf("Data transferred: %.2f MB\n", data_mb);
        printf("Communication efficiency: %.2f MB/s\n", efficiency);
    }

    return 0;
}

```

```

jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ pwd
/home/jella_pragathi/OSSP
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ nano producer_consumer.c
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ gcc -Wall -Wextra -g producer_consumer.c -o producer_consumer
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./producer_consumer

Consumer Process
Child PID: 899
Items consumed: 1000000

Producer Process
Parent PID: 898
Items produced: 1000000
Communication time: 0.257368 seconds
Data transferred: 3.81 MB
Communication efficiency: 14.82 MB/s
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ls -l producer_consumer.c producer_consumer
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19280 Aug 31 03:41 producer_consumer
-rw-r--r-- 1 jella_pragathi jella_pragathi 1865 Aug 31 03:41 producer_consumer.c

```

Task 2:

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

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GNU nano 8.7.1

pipeline.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    // First child: executes ls -l
    pid_t pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid1 == 0)
    {
        close(pipefd[0]);

        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(1);
    }
}
```

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```
jella_pragathi@LAPTOP-RQQ( x + v - □ ×
GNU nano 8.7.1 pipeline.c *
    exit(1);
}

// Second child: executes grep ".c"
pid_t pid2 = fork();

if (pid2 < 0)
{
    perror("fork");
    return 1;
}

if (pid2 == 0)
{
    close(pipefd[1]);

    // Redirect stdin from pipe
    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);

    execlp("grep", "grep", ".c", (char *)NULL);

    perror("execlp grep");
    exit(1);
}

// Parent process
close(pipefd[0]);
close(pipefd[1]);

waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

printf("\nPipeline execution completed.\n");

return 0;
```

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^X Exit	^R Read File	^I Replace	^U Paste	^J Justify

```

jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ls -l | grep ".c"
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:01 docs
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:01 include
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19280 Aug 31 03:41 producer_consumer
-rw-r--r-- 1 jella_pragathi jella_pragathi 1865 Aug 31 03:41 producer_consumer.c
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:02 scripts
-rw-r--r-- 1 jella_pragathi jella_pragathi 1507 Aug 23 16:14 session4_wait.c
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 src
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16224 Aug 26 03:26 task1_escape
-rw-r--r-- 1 jella_pragathi jella_pragathi 572 Aug 26 03:25 task1_escape.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16296 Aug 26 03:30 task2_child
-rw-r--r-- 1 jella_pragathi jella_pragathi 807 Aug 26 03:29 task2_child.c
-rw-r--r-- 1 jella_pragathi jella_pragathi 780 Aug 23 16:41 zombie.c
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ nano pipeline.c
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ gcc -Wall -Wextra -g pipeline.c -o pipeline
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./pipeline
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:01 docs
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:01 include
-rw-r--r-- 1 jella_pragathi jella_pragathi 1181 Aug 31 03:46 pipeline.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19280 Aug 31 03:41 producer_consumer
-rw-r--r-- 1 jella_pragathi jella_pragathi 1865 Aug 31 03:41 producer_consumer.c
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug  1 08:02 scripts
-rw-r--r-- 1 jella_pragathi jella_pragathi 1507 Aug 23 16:14 session4_wait.c
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 src
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16224 Aug 26 03:26 task1_escape
-rw-r--r-- 1 jella_pragathi jella_pragathi 572 Aug 26 03:25 task1_escape.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16296 Aug 26 03:30 task2_child
-rw-r--r-- 1 jella_pragathi jella_pragathi 807 Aug 26 03:29 task2_child.c
-rw-r--r-- 1 jella_pragathi jella_pragathi 780 Aug 23 16:41 zombie.c

Pipeline execution completed.

```